Dept. of Statistics

Islamic University, Kushtia-7003

Course: CSE-1106 (Computer Fundamentals and Task Oriented Software)

Lecture: 01

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- **Basic Concept of a Computer System**

The word “Computer” comes from the word “Compute” which means “to calculate”. Hence people usually consider a computer to be a calculating device that can perform arithmetic operations at a high speed. Although the original objective of inventing a computer was to create a fast calculating device, we now define a computer as a device that operates upon data because more than 80% of work done by today’s computers is data processing. Data can be anything like bio-data of applicants when computer is used for short listing candidates for recruiting; marks obtained by students in various subjects when used for preparing results; details (name, age, sex, etc.) of passengers when used for making airline or railway reservations; or number of different parameters when used for solving scientific research problems, etc. It should be noted from the above examples that, data can be either numerical, non-numerical, or a mixture of both.

A computer is often referred to as a data processor because it is an electronic machine that can store, process, and retrieve data whenever desired. It can only carry out instructions given to it. A set of instructions that directs its actions is called a program. Different programs are used to solve different problems. The activity of processing data using a computer is called ***data processing***. Data processing consists of three sub-activities: capturing input data, manipulating the data, and managing output results. When raw data is processed into a meaningful format or arranged in an order and format that is useful to people, then we say this information. Hence, data is raw material used as input to data processing and information is processed data obtained as output of data processing (Figure -1.1).

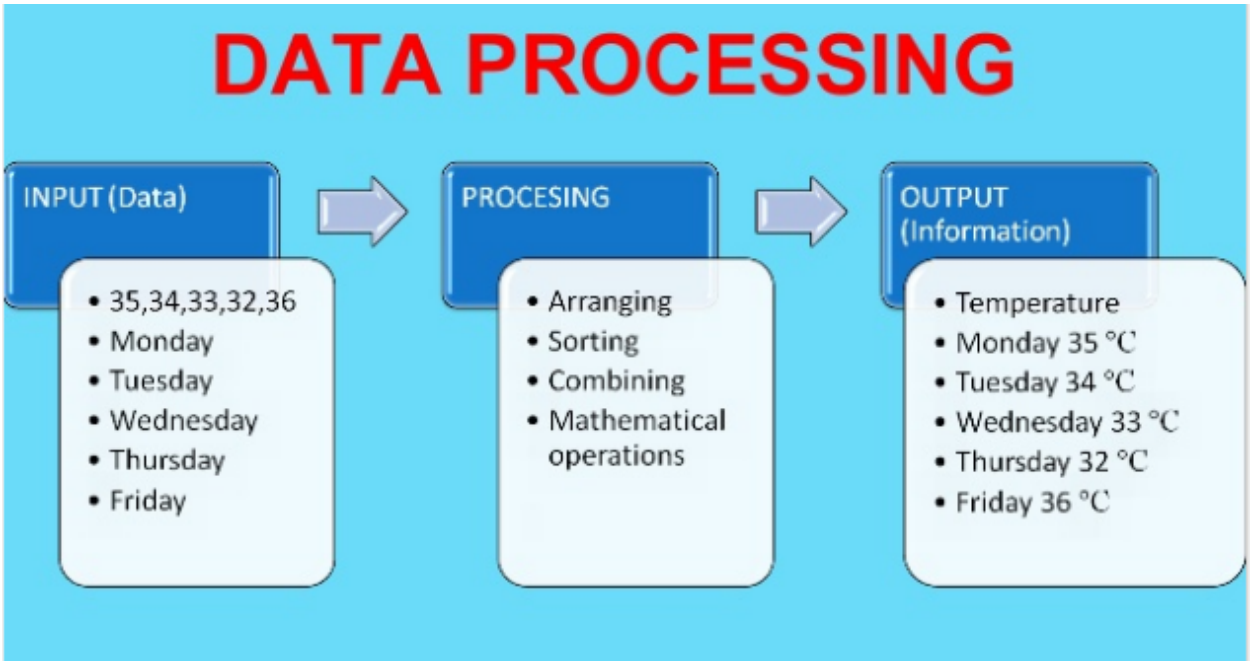

Figure-1.1: A computer (also as a data processor) converts data into information

- **Characteristics of Computers**

Increasing popularity of computers has proved that is a very powerful and useful tool. The power and usefulness of this popular tool are mainly due to its following characteristics:

1. **Automatic:** An automatic machine works by itself without human intervention. Computers are automatic machines because once started on a job, they carry out the job (normally without any human assistance) until it is finished. However, computers being machines cannot start themselves and cannot go out and find their own problems and solutions. We need to instruct a computer using coded instructions that specify exactly how it will do a particular job.

2. **Speed:** A Computer is a very fast device. It can perform in a few seconds, the amount of work that a human being can do in an entire year – if he/she worked day and night and did nothing else. In other words, a computer can do in a few minutes what would take a man his entire lifetime. A powerful computer is capable of performing several billion (10^9), even trillion (10^{12}), simple arithmetic operations per seconds.

3. **Accuracy:** In addition to being very fast, computers are very accurate. Accuracy of

computer is consistently high and the degree of its accuracy depends upon its design. However, error can occur in a computer. These errors are mainly due to human rather than technological weaknesses. For example, errors may occur due to imprecise/inexact thinking by a programmer (a person who writes instructions for a computer to solve a particular problem) or incorrect input data.

4. **Diligence:** Unlike human beings, a computer is free from monotony/indifference, tiredness, and lack of concentration. In can continuously work for hours without creating any error and without grumbling.

5. **Versatility:** Versatility is one of the most wonderful things about a computer. A computer is capable of performing almost any task, (e.g. preparing results of an examination, preparing electricity bills, helping office secretary, helping teachers and students in educational matters, etc.) if the task can be reduced to a finite series of logical steps.

6. **Power of Remembering:** A Computer can store and recall any amount of information because of its secondary storage (a type of detachable memory) capability. It can retain a piece of information as long as a user desires and the user can recall the information whenever required.

7. **No IQ:** A computer is not a magical device. It possesses no intelligence of its own. Its IQ is zero, at least until today. It has to be told what to do and in what sequence. Hence, only a user determines what tasks a computer will perform. A computer cannot take its own decision in this regard.

8. **No Feeling:** Computers are devoid of emotions. They have no feelings and no instincts/motivations because they are machines. Although men have succeeded in building a memory for computer, but no computer possesses the equivalent of a human heart and soul. Based on our feelings taste, knowledge, and experience we often make certain judgements in our day-to-day life whereas, computers cannot make such judgements on their own. They make judgements based on the instructions given to them in the form of programs that are written by us (human beings).

- **Limitations / Disadvantages of Computer:**

  - Computers can not think.
  - Computers can not do anything without human instructions.
  - Computers can not make any adjustment as human being can do